S(18): DYADIC REDUCTION AND RELATIVE FORMAL IMMERSIONS

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Theorem. For elliptic curves over number fields of exact degree 18,
 S(18) = {2,3,5,7,11,13,17,19,23,29,31,37,41,43}.
This is a computer-assisted proof. The complete prime and support coverage has passed, with no unresolved case.

1. OCCURRENCES AND EXACT DEGREE

Published sextic examples supply every prime through19; extending those fields by degree3 gives exact degree18. The script verify_extension_occurrences.m independently verifies explicit an irreducible degree18 example for23, degree9 examples for29 and31, and a degree6 example for37. The latter three fields extend to degree18. The script verify_occurrences.m checks explicit irreducible degree18 fields for41 and43. In every case it verifies nonsingularity and the exact prime order of (0,0) on the associated Tate normal form. These data are retained in extension_occurrences.json and occurrences.json. Completeness of the place search used to find the examples is not needed.

For an upper bound no classification in degree9 is needed. A point on X_1(p) coming from a degree18 field has some exact degree d dividing 18. Its trace over the degree18 field is D=(18/d)*D0, where D0 is an irreducible noncuspidal rational divisor of degree d. We work with this effective degree18 trace. It has no cusp in its support. If a diamond subgroup preserves D, it preserves D0; its geometric orbits on D0 have a common length dividing d, hence dividing18. This is the only irreducibility consequence used in the argument.

2. THE IMPROVED DYADIC BOUND

Use the previously proved dyadic winding-quotient theorem, including its integral cotangent-lattice assertion 2M subset L subset M. We do not replace this inclusion by equality. With m=36=2*18 and auxiliary modulus181, all180 matrices
 epsilon(na)-epsilon(nu/a), 1<=n<=36, 1<=a<=180,
have rank36 over F_2; epsilon indicates the upper half of nonzero residues modulo181. The intersection formula applies for
 p > 2*36*181 = 13032.
The doubled-divisor argument in the formal neighborhood modulo4 therefore excludes traces supported at a single rational cusp of X_0(p). Doubling accounts for the possible factor2 lattice defect and for the exponent2 kernel of reduction on rational dyadic torsion. The matrices are retained and independently rebuilt in Sage.
The coefficient pairing identifies the saturated winding-form lattice with the dual of the winding Hecke module. Multiplying a putative relation by T_2-3 and using the integral DKSS intersection formula contradicts these ranks. Thus the cotangent image contains2*Z_2^36. In local coordinates the target functions can be chosen as2*x_i plus terms of order at least2. For x in4*Z_2^36, a nonzero coordinate of minimum valuation k>=2 gives a linear term of valuation k+1<2k, so the map has no nontrivial zero there. If E_D(t)=1+2B(t) is the elementary symmetric polynomial of a divisor reducing to a multiple of the cusp, then E_(2D)(t)=1+4(B(t)+B(t)^2). The doubled divisor lies in the neighborhood modulo4, and its winding-quotient image is zero because the original torsion image reduces to zero and is killed by2. This proves the single-cusp exclusion, including ramified divisors.

Hasse gives (2^9+1)^2=263169 for good reduction in residue degree at most18. If p>263169, no good-reduction place can support the torsion; additive reduction and the identity component of multiplicative reduction are also impossible. At every dyadic place the point must then meet a nontrivial multiplicative component, so its cyclic subgroup has the same cusp specialization on X_0(p). The dyadic theorem excludes this. Thus p<=263169.

We also use parity. For p>100000, ordinary reduction in residue degree at most17 is impossible, since its group order is even and at most 131797<2p. Supersingular orders in degree17 are
 130561=137*953, 131073=3*43691, 131585=5*26317,
and smaller residue degrees have Hasse bounds below100000. In degree18, ordinary reduction requires #E(F_(2^18))=2p, so
 261121 <= 2p <= 263169.
The complete supersingular twist orders in degree18 are
 261121,261633,262145,262657,263169,
with repetitions in the twist list. Magma checks them directly. Their factorizations leave just262657 above100000. Checking 2^f+/-1 for 1<=f<=18 adds only131071, already in the ordinary interval.

Consequently, above100000 there are exactly81 admissible primes: 80 in the short interval130561..131584 and262657. Every other prime in the dyadic range reduces to the already excluded single-cusp case.

3. GLOBAL HECKE EXCLUSIONS AND THE RANGE OF THE CHARACTER TABLE

For primes below100000, use Derickx--Stoll, Part I: Asymptotics, Table2, Proposition4.1, Remark4.4, and Corollary7.5, applied to d|18. If a subgroup order does not divide18, it does not divide d, and each size inequality checked at18 also holds at d. The exceptional CM possibilities for even d|18 are5,7,13,19,37,55, so they contribute no prime above43. The ell=2 instances use only levels<=157, where the required rational2-torsion assertion is available; ell=3 is used when necessary. All inequalities are evaluated exactly.

These criteria leave exactly
 47,53,59,61,67,71,73,79,83,89,97,101,103,107,109,113,127,131.
No character-table assertion is extrapolated beyond100000.

For each of the81 larger admissible primes p, put m=(p-1)/2. Choose r with v=v_r(m)>w=v_r(18). Screen all character orders n|m satisfying
 r^(v-w) | n.
If all these orders are non-strange, the valuation of str(p) at r is at most v-w-1. Hence
 v_r((p-1)/(2*str(p))) >= w+1,
so (p-1)/(2*str(p)) does not divide18. The size inequality in Corollary7.5 holds throughout this large range, and excludes p. There are615 selected orders in total. Exact finite-field modular symbols and cyclic-span calculations certify their non-strangeness; random choices only search for exact rank witnesses. Failed memory-limited attempts are not used as exclusions. The high-memory prime 262657 is checked separately. The selected orders, code, seeds, and completed per-character logs are retained.

4. THE LOCAL CENSUS AND PRINCIPAL-DIVISOR OBSTRUCTIONS

At each remaining level, the finite-field census enumerates every

j-invariant of each degree<=18, every twist, and every cyclic line in E[p]. It retains all closed-point diamond orbits, with their exact closed degrees. Nonrational cusps have degree ord_p(2) modulo sign. Every effective support multiset of total degree<=18, including repeated closed points, is included in the support ledger.

For ordinary points use actual elliptic3-isogenies, including all kernel factors and trace multiplicities of the target closed points. Frozen canonical q-expansion bases identify points and local jets via two modular units. The new level131 has genus651, q precision3300, and unit pole degrees225 and235. The local vanishing orders used to prove its unit relations exceed the global section degree bounds. Multiplication tensors are checked beyond Sturm bounds.

For an ordinary source divisor Z, apply alpha*A_3, where
 A_3 = T_3 - 1 - 3*diamond3,
 alpha = sigma6-1 at61,
         sigma12-1 at97,
         (sigma-1)(sigma5+1) at101,
         1 at47,59,71,
         sigma-1 otherwise.
Here sigma is the recorded diamond generator. The character table proves finite image on rational Jacobian points; A_3 kills rational torsion and the rational cusp contribution. The earlier level97 exceptions use the valid alternative (sigma-1)*Phi12(sigma), not the superseded invalid polynomial. Their corrected degree17 certificates are inherited through explicit Tate-normal-form orbit identifications.

If U-V=alpha*A_3(Z) were principal, choose a nonzero
 f in H^0(L(-U)). Then there would be a nonzero g in H^0(L(-V)) with
 h*g in f*H^0(L) for every h in H^0(L(-U)).
Here L=K^2 at47,53,61, and L=K elsewhere for the new calculations. The displayed necessary conditions are linear in g. Stored exact matrices have rank dim H^0(L), proving nonprincipality.
Only a few h are needed when they already give full rank. The tensors, sections f and h, jet conditions, and rank matrices are retained. Every requested jet multiplicity is asserted to be within the actual retained precision. Native matrix operations were replayed against the earlier coefficient-by-coefficient implementation with identical jets.

The complete ordinary degree<=17 certificates are inherited after matching each actual Tate-normal-form diamond orbit, rather than assuming random finite-field enumerations use identical coordinates. All new degree18 ordinary orbits are checked. New pairs exhaust total 18 and every relative diamond position. At the newly studied level131, all pairs with totals16 and17 are also checked, with their cusp complements. Earlier pairs with totals14..17 at the other levels remain valid with arbitrary rational cusp complements.
New mixed obstructions cover73 with SS6+O12,113 with SS7+O11, and127 with NC7+O11. Earlier mixed obstructions cover the smaller totals.

5. THE INTEGRAL RELATIVE LEMMA

Let S be smooth of relative dimension d over Z_2, and let L subset S be a smooth closed formal family of dimension r through a special-fiber point. Suppose an actual integral map Phi:S->A to a smooth group scheme vanishes on L and has cotangent rank d-r there. Its completed zero fiber equals L: choose normal coordinates v for L and d-r target functions with independent normal differentials. Since these functions

vanish on L, they equal an invertible matrix times v in the completed local ring. Their common zero locus is exactly L.

This statement uses no integration denominator bound at2. The maps used here are odd-prime torsion annihilators composed with h-1 for suitable diamonds h. They kill rational Jacobian points and quotient pullback Jacobians. Fixed rational cusp divisors contribute zero. A fixed local h-invariant divisor can also be included, since its image under h-1 is zero and the remaining base-point term is cuspidal.

After Fricke transport, cotangent matrices in the q-expansion basis are (D_h-1)(T_q-D_q-q), up to invertible diamond factors. The same map and basis are used on all support and cusp columns. Coefficient-coordinate matrices at closed points differ from trace-pairing matrices by invertible changes of field basis. The new relative matrices are also rebuilt independently in Sage.

6. RELATIVE RESIDUE DISKS

At53, H=<diamond4> has order13 and acts freely on SS13. The local family is F5+pi^*Y, with F5 a fixed rational cusp divisor of degree5. The normal rank is17 for all142506 cusp patterns. Thus a rational D in such a disk contains F5, contrary to its noncuspidal support.

At61, H=<diamond3> has order5 and acts freely on SS15. The family is F3+pi^*Y^(3). The normal rank is15 for all4960 cusp patterns, again forcing a nonzero rational cusp contribution.

At73, h=diamond2 has order9. Its order3 subgroup fixes the geometric points of SS6 and has nontrivial tame tangent action. Their unique fixed-locus lifts form an h-invariant local divisor Z0 of degree6. The36 rational cusps form four h-orbits of size9. Extract full fibers from the degree12 rational cusp complement: C=R+pi^*E. Then deg(E) is0 or1 and deg(R)=12-9*deg(E)>0. The family

 Z0+R+pi^*Y^(deg(E))

has exactly the predicted tangent kernel. All2843832291 normalized cusp patterns for each of the four SS translates pass the rank check. D would contain R, a contradiction.

There is a second relative case at73: NC9+C9. NC9 is an unramified H-fiber. Its base cotangent rank is8; all47929363 normalized cusp patterns per translate have exactly the additional relations from complete rational cusp fibers. If a fixed cusp part remains, D has a cusp in its support. If none remains, D is the pullback of a rational effective divisor of degree2 on
X_1(73)/H. Section7 excludes this alternative.

At127, the difficult mixed case is the exceptional O9+NC7+C2. Use the product of the order9 and order7 diamond differences, composed with T_2 and T_3 torsion annihilators; the level127 rational cuspidal torsion result justifies T_2. The two quotient pullbacks give a smooth local family of dimension2. The base rank is14 and all2016 two-cusp patterns have normal rank16. D would contain its fixed degree2 cusp divisor. The coprime diamond orbit lengths7 and9 normalize both nonrational supports simultaneously, so all relative positions are included.

At103 and131, quotient cusp tests have one source of deficient patterns: a full quotient cusp fiber of degree17 or13, respectively. All other patterns pass absolute formal immersion. The complete

relation spaces show these are the only deficient patterns. The map (diamond(root)-1)(T_3-diamond3-3) on the intermediate curve kills the pullback of J_0. Its intrinsic quotient-cotangent rank is17 at every full fiber plus fixed cusp divisor of degree1 or5. There are17 and6188 such patterns. Thus a rational noncuspidal divisor would contain a fixed cusp contribution. This argument applies to the effective pushforward trace even if the point degree drops on the quotient.

7. QUADRATIC POINTS ON THE ORDER9 QUOTIENTS

The new order9-invariant degree18 divisors at73 and127 require excluding quadratic points on Y=X_1(p)/H9. A degree2 effective divisor on either quotient cannot have rational noncuspidal support, since it would map to X_0(p)(Q). Thus it is an irreducible quadratic point, whose image on X_0(p) is also quadratic. This also rules out a doubled rational quotient point when D itself is a multiple trace.

Use Box, Quadratic points on modular curves with infinite Mordell--Weil group, section4.7 (the completed quadratic-point list at73), and Adzaga, Keller, Michaud-Jacobs and Vukorepa, Table16 for127. At73 the list has three non-CM quadratic j-invariants and eight rational CM j-invariants. At127 all seven listed j-invariants are rational CM values. The exact values and quadratic fields are recorded in quotient_quadratic.m. For the -12288000 entry we check both Q(sqrt(-1)) and Q(sqrt(-3)), so the field label in that table is not an unverified assumption.

For each listed point choose a rational prime q split in its quadratic field and an explicit root theta modulo q. Let F be a Frobenius matrix on a reduced elliptic curve with the required j-invariant. If a point lifted to Y, an H9-orbit modulo automorphisms would be Frobenius-stable. Consequently det(F^e-1)=0 mod p, with e=18, or e=36 for j=1728. The chosen primes give nonzero determinants in every case. At CM points the split-prime choice makes Frobenius commute with the automorphisms; at other points the exponent18 removes the quadratic-twist ambiguity. All determinants are independently recomputed from elliptic-curve point counts in Sage.

Separate Frobenius certificates exclude degree18 branch CM points j=0 at73,109,127 and j=1728 at97,109,113. If an automorphism orbit of a nonzero p-torsion vector had degree dividing18, F^(18*w) would fix a nonzero vector, where w=6 or4. Each certificate gives a nonzero determinant.

8. STABILIZER CASES AND COMPLETENESS

If h fixes the nonrational reduction and C is the rational cusp lift, finite image implies y=(h-1)[D-C] is torsion reducing to0, so2y=0. The known lower bound

$$\text{gon}_{\mathbb{Q}}(X_1(p)) \geq 325(p^2-1)/65536$$

then proves hD=D and hC=C whenever twice(18+deg(C)) is below it. At73, actual rational2-torsion is fixed by diamond2. The norm relation for the odd order9 action therefore gives y=0, permitting the stronger bound18+deg(C). At89 the previously verified odd-degree Hecke annihilator on both anti-Jacobian Betti signs likewise gives y=0.

At73 this handles SS12+C6, SS6+NC9+C3, and all degree18 nonrational supports, including SS18, three SS6 points, or two NC9 points. Either C cannot be invariant, D descends to the excluded quadratic quotient, or D consists of the excluded j=0 branch points.

At89 the order11 action cannot have noncuspidal orbits dividing18.
At97 the order4 action can only give the excluded j=1728 branch case.
At109 the order18 action handles SS9+C9, SS12+C6, two SS9 points, and the newly possible NC18. A free invariant D descends to a rational point of the order18 quotient, hence to X_0(109)(Q), which is impossible by Mazur; both possible CM branch cases are also excluded.
At113 the order14 action has orbit lengths14 or7, neither dividing18.
The exceptional O8 and repeated O8 cases use order8, whose possible orbit lengths8 and4 also do not divide18. In SS7+O8+C3 the involution forces invariance; a free pair has even specialization multiplicity at an involution-fixed point, whereas SS7 has multiplicity1. Its only remaining alternative is the excluded j=1728 branch case.
At127 the order7 action handles pure SS14 and NC7 or repeated NC7 supports. The exceptional O9 and repeated exceptional O9 cases use order9 and descend to the excluded quadratic quotient, unless they are the separately excluded j=0 branch case.

The support checker enumerates all284 support-degree multisets. Every ordinary orbit, relative pair position, cusp multiplicity pattern, CM case, and prime in the full dyadic range is assigned a completed certificate or a proved mathematical exclusion in final_coverage.json.
All checks have passed:23079 primes,284 support types,615 selected character orders, and11854 new independent obstruction matrices.
The285-file inherited proof manifest also matches without change.
Together with the occurrence certificates, this proves the theorem.

REFERENCES AND RETAINED PROOF INPUTS

Derickx--Stoll, Prime order torsion on elliptic curves over number fields, Part I: Asymptotics, Theorem1.2, Theorem3.3, Proposition4.1, Remark4.4, Table2, and Corollary7.5:
https://www.mathe2.uni-bayreuth.de/stoll/papers/asymptotics.pdf

Box, Quadratic points on modular curves with infinite Mordell--Weil group, section4.7, completed list:
https://arxiv.org/html/1906.05206

Adzaga--Keller--Michaud-Jacobs--Vukorepa, Computing quadratic points on modular curves X_0(N), Table16:
https://arxiv.org/html/2303.12566v2

The earlier dyadic lattice argument and the corrected inherited local certificates are retained in outputs/S13_proof.pdf and work/degree17/PROOF.txt. The latter’s285-file proof manifest has been checked against the actual inherited files. In particular, only the corrected level97 operators are used.

The final machine-readable ledgers are final_coverage.json and support_coverage.json. independent_rank_checks.json and independent_rank_low131.json certify all11854 new obstruction matrices. The relative-map replays and character-order coverage have their own verified JSON files. REPLAY.txt gives exact verification commands; proof_manifest.json gives SHA256 hashes of the retained proof inputs.